

OVERVIEW OF THE PROCESSING

Controller(s)

Field	Value
Controller	Customer operating a private video streaming platform using shystream-v1
Management units responsible	Product Engineering; Legal / Compliance; Data Protection
DPO or similar function	Customer DPO or privacy contact

Processor(s) and Sub-processor(s)

#	Processor	Obligations and Tasks
1	co-mission LLC / Shyware	Provides SDK, shystream client, and optional hosted API under the Shyware
2	Cloud infrastructure provider	Hosts canonical node, API, and database; processes personal data under A
3	Identity verification provider	Biometric identity verification for identity-gated stream access and recovery
4	Video infrastructure provider	CDN and origin serving for sealed video segments; receives sealed segment

Name of the Processing

Internal name: Shyware Private Video Streaming / shystream-v1 DPIA

Repository evidence: Consumers/MUX_COM; shyware/sdk/web/clients/utilities/streamClient.js; patent/mainhub/dpia/evidence_index.md.

DPIA Technical Sheet

Field	Value
Current version	0.1 draft
Reason to conduct DPIA	High-risk processing: viewer identity and viewing behavior are both sensitive; correl
Scope	Private video streaming using shystream-v1 sealed segment events; viewer identity e
Publication	Internal domain DPIA; customer-specific adaptation required before production use

SYSTEMATIC DESCRIPTION OF THE PROCESSING

High-level Description of the Processing

Processed Personal Data

#	Personal Data Item	Data Type
1	Stream event identifier (List 1 record)	Direction-free opaque identifier; no viewer identity
2	Sealed segment payload (List 1 record)	AES-256-GCM sealed clip metadata; content inaccessible without key
3	Identity hash (List 2 record)	SHA-256 of verified viewer identity; no clip identifier or direction
4	Aggregate view count per scoping period	Integer count of List 1 and List 2 entries; no individual linkage
5	Period-close attestation	HSM-signed Merkle roots over disjoint List 1 and List 2 sets
6	Off-chain reconciling record	Per-viewer stream history; accessible only via identity-gated reconciliation

Canonical state architecture: Items 1–5 are canonical ledger records. No record in canonical state associates any List 1 stream event with any List 2 viewer identity hash. The rejection predicate refuses any state transition that would create such an association. Viewer identity is structurally absent from canonical clip-view records — not by policy, by write architecture.

Purposes of the Processing

#	Purpose	Personal Data	Legal Basis
1	Private video streaming — aggregate view tracking	Items 1, 2, 4, 5	Art. 6(1)(f) legitimate interests; identity
2	Identity verification — sybil and access control	Item 3	Art. 6(1)(b) / (f) service integrity; c
3	Period-close attestation — compliance audit	Item 5	Art. 6(1)(f) / (c) aggregate only; no
4	Viewer-authenticated stream history retrieval	Item 6	Art. 6(1)(b) contract performance; v

Authority Partition This is a sealer-governed embodiment (no unique-person count-match).
Three parties:

Authority	Role
Auth (Cognito / Firebase)	Account layer; authentication gate for all API calls including erasure requests
IDV/KYC (Didit)	Verifies account belongs to a real, verified viewer; assessed on authenticated users
Reconciling authority	Off-chain linkage store + sealer keys

ANALYSIS OF THE PROCESSING

Lawfulness of the Processing

Customer controllers must define the lawful basis per deployment. The primary basis for aggregate view-event processing is Art. 6(1)(f) legitimate interests, supported by the structural non-linkability of viewer identity to clip-view direction in canonical state. Where personalized streaming history or viewer profiling occurs through the off-chain reconciling store, the legal basis for that secondary processing must be separately assessed.

Data Minimisation and Retention

Data Item	Minimisation Position
Sealed segment payload	Store only metadata necessary for entitlement verification; minimize content fingerprinting
Identity hash	Scope to product and scoping period
Off-chain reconciling record	Disable unless viewer history retrieval is required
Video CDN logs	Exclude viewer identity from CDN request logs where possible

RISK ASSESSMENT

Threats

#	Threat	Materialisation
1	Viewer-content correlation via CDN logs	CDN provider retains per-request logs linking IP to content segments
2	Sealer key disclosure	Reconciling authority discloses or loses sealer keys; sealed payload is compromised
3	Cross-scoping-period viewer linkage	Identity hash reused across scoping periods enabling long-term viewer identification
4	Video segment fingerprinting	Third party correlates content segments to viewer via traffic analysis
5	Deletion failure	Canonical/off-chain/CDN stores diverge on erasure

Residual Risk Assessment

Risk	Additional Measures
CDN log correlation	CDN DPA with log minimization; sealed segment requests without viewer identity in logs
Sealer key disclosure	Dual-control sealer access; audit logs; WORM backup on separate provider
Cross-period linkage	Scope identity hashes per product, scoping period, and viewer
Segment fingerprinting	Segment obfuscation; avoid predictable segment naming
Deletion failure	End-to-end deletion evidence: canonical, off-chain, CDN

Operator Compliance Instructions

- **Art. 6 lawfulness basis** — Identify and document the legal basis for each processing purpose before production. Aggregate view-event processing: Art. 6(1)(f). Personalized history: assess separately.
- **Art. 28 sub-processor DPAs** — Execute signed DPAs with cloud infrastructure, IDV provider, and video CDN provider before processing personal data.
- **CDN log minimization** — Configure the video CDN provider to minimize retention of per-request logs that could link viewer identity to content segments.
- **Art. 30 Records of Processing** — Generate from shyconfig using `generate-rop.mjs`; complete all `REQUIRED_OPERATOR_INPUT` fields.

- **Storage limitation (Art. 5(1)(e))** — Add `retention` block to `shyconfig.json`; configure and enforce deletion cron for sealed segment records and off-chain linkage.
- **Art. 17 erasure path** — Erasure = sealer key deletion by reconciling authority on authenticated viewer request; canonical L1/L2 records persist (BFT-immutable); payload becomes permanently undecryptable. Ensure CDN-layer segment references are also purged.

CONCLUSION AND DECISION

Draft decision: Conditional approval. The shystream-v1 canonical state architecture achieves structural non-linkability between viewer identity and clip-view events. The residual risks lie in CDN log correlation, sealer key governance, and erasure completeness across the CDN layer. Production deployment requires: CDN DPA with log minimization, dual-control sealer key access, scoping-period-scoped identity hashes, and end-to-end deletion evidence covering canonical, off-chain, and CDN stores.